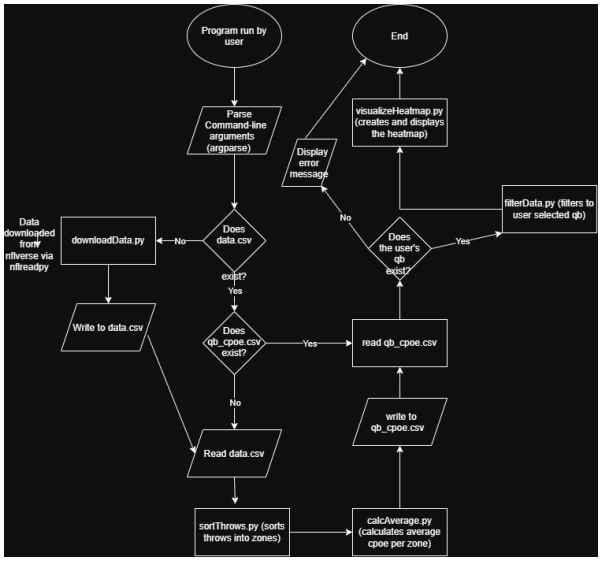
Problem Statement:
When I was watching the NFL games today, I noticed that quarterbacks vary significantly to the naked eye on accuracy depending on what part of the field they are throwing to. To confirm this theory, I want to build a QB heatmap tool that lets the user select any QB and see how accurate he is in each zone of the field (short, middle, long) (left, middle, right). I care about this because I want to be able to know QB tendencies in order to accurately judge how good a QB truly is overall.
Solution:
I am going to build a tool that allows the user to select a NFL QB, which then displays his heatmap (how accurate he is in each portion of the field, using CPOE (Completion percentage over expected)). For the MVP, this will be in the command line, but this may change after the MVP is completed to add features like a GUI or something else.
Tech Stack for the MVP:
Language: Python
Data: Source play by play CPOE data from nflverse using nflreadpy (subject to change if I find a better source).
Libraries: Pandas, matplotlib, seaborn, numpy (maybe)
The libraries needed may change with changes to the project
Data will be stored in a CSV file after processing (each zone's stats will be saved to it). Columns that will be used are: name, location (left, middle, right), depth (short, mid, deep), and cpoe
Milestones:
1. Set up repo and environment
2. Load play-by-play data from nflverse
4. Sort each throw into zones and calculate average CPOE for each zone
5. Store average CPOE data in a .csv file
6. Create and visualize the heatmap
7. Set up command-line arguments to allow the user to pick the QB



https://github.com/brgilly16/NFL-QB-Heatmaps